

COMPUTER REPAIR AND MAINTENANCE

PRACTICAL ASSESSMENT 2

Unit of Competency: PERFORM COMPUTER REPAIR AND MAINTAINANCE

Unit code: UNIT CODE: 0714 351 04A

ELEMENT3: Test Computer Component Functionality

Total Marks: 50

Duration: 2 Hours

Name.....Date.....

Regno.....sign.....

- ✓ *Instructions*
- ✓ *Read all tasks carefully before starting.*
- ✓ *Observe all safety precautions when handling computer equipment.*
- ✓ *Record all observations and test results.*
- ✓ *Complete all tasks and submit the required report.*

SCENARIO

You are working as a Computer Repair and Maintenance Technician in a computer service center. A customer has brought a desktop computer that is experiencing startup and performance problems. The customer reports that the computer occasionally fails to boot, freezes during operation, and some peripherals are not functioning correctly.

Your supervisor has assigned you the responsibility of testing the computer, evaluating its condition, verifying repaired or replaced components, and preparing a functionality report.

Equipment's provided

- Desktop computer (faulty)
- Monitor
- Keyboard and mouse
- Diagnostic software tools
- Network cable
- Screwdriver set
- Replacement component(s)

TASK 1: PERFORM POWER-ON SELF-TEST (POST)

- a) Power on the computer and observe the startup process.
- b) Record all POST messages, beep codes, or warning messages displayed during startup.
- c) Identify the hardware components detected during POST.
- d) Determine whether the computer successfully completes POST.
- e) Record your findings.

TASK 2: PERFORM COMPUTER COMPONENT TESTS

Test the functionality of the following computer components and record the results:

- Memory (RAM) test using Memtest86
- Storage device (HDD/SSD) test using crystal disk health info
- Keyboard
- Mouse
- Monitor

For each component:

- a) State the testing method used.
- b) Perform the test.
- c) Record the observations.
- d) State whether the component is functional or faulty.

TASK 3: TEST REPAIRED OR REPLACED COMPONENTS

A repaired or replacement component will be provided.

- a) Identify the repaired or replaced component.
- b) Install or reconnect the component correctly.
- c) Perform appropriate tests to verify functionality.

- d) Confirm whether the fault has been corrected.
- e) Record the test results.

TASK 4: EVALUATE TEST RESULTS

Using the results obtained from the tests performed:

- a) Analyze the results.
- b) Identify any faulty components.
- c) Determine possible causes of the faults.
- d) Recommend corrective actions.
- e) State whether the computer is fit for normal operation.

TASK 5: GENERATE A COMPUTER COMPONENT FUNCTIONALITY REPORT

Prepare a professional report containing the following:

1. Candidate Name
2. Admission Number
3. Date of Assessment
4. Description of the reported problem
5. POST observations
6. Components tested
7. Testing techniques used
8. Test results obtained
9. Repaired or replaced components tested
10. Evaluation of results
11. Recommendations
12. Conclusion